EE4402 Project

Students：

Huang Shaojie U2401025

Kan Ziyang U2401026

Li Wushuai U2401027

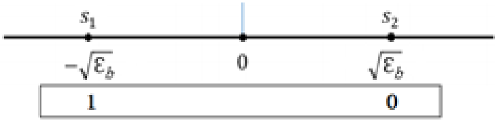
Li Linyangyang U2401082

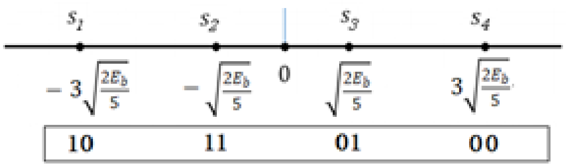
(Our simulation is based on matlab)

Task 1

Assume all symbols are equiprobable. Denote the average bit energy by E_b. Derive and simulate the bit error rate (BER) for PAM when the Gray mapping is applied.

a. If 2-PAM and 4-PAM have the same average bit energy E_b, indicate the symbols in vector form for 2-PAM and 4-PAM using E_b.





Assume $S_{ml}(t) = A_m p(t) \qquad A_m = 2m - 1 - M \qquad 1 \leq m \leq M = 2^k$

Denote $\varepsilon_p = \int_{-\infty}^{\infty} p(t)^2 dt \quad \rightarrow \quad \varepsilon_{ml} = {A_m}^2 \varepsilon_p = E_b$

PAM has dimension one $\Phi_l(t) = \frac{p(t)}{\sqrt{\varepsilon_p}}$

2PAM(M=2):

$\varepsilon_{ml} = {A_m}^2 \varepsilon_p = \varepsilon_p$

$\varepsilon_{avgl} = \varepsilon_p \qquad \varepsilon_{bavgl} = \frac{\varepsilon_{avgl}}{\log_2 M} = \varepsilon_p = E_b$

$S_1 = \langle S_{1l}(t), \Phi_l(t) \rangle = -\sqrt{\varepsilon_b} = -\sqrt{E_b} \quad S_2 = \langle S_{2l}(t), \Phi_l(t) \rangle = \sqrt{E_b}$

Vector form: $S_1 = (-\sqrt{E_b}, 0) \;\; S_2 = (\sqrt{E_b}, 0)$

4PAM(M=4):

$A = \pm 1, \pm 3 \qquad \varepsilon_{ml} = {A_m}^2 \varepsilon_p = 1,9,9,1\ \varepsilon_p$

$\varepsilon_{avgl} = 5\varepsilon_p \qquad \varepsilon_{bavgl} = \frac{\varepsilon_{avgl}}{\log_2 M} = \frac{5}{2}\varepsilon_p = E_b$

$$S_1 = \langle S_{1l}(t), \Phi_l(t) \rangle = \int_{-\infty}^{\infty} -3p(t) * \frac{p(t)}{\sqrt{\varepsilon_p}} dt = -3\sqrt{\varepsilon_p} = -3\sqrt{\frac{2E_b}{5}}$$

$$S_2 = \langle S_{2l}(t), \Phi_l(t) \rangle = -\sqrt{\frac{2E_b}{5}}$$

$S_3 = \langle S_{3l}(t), \Phi_l(t) \rangle = \sqrt{\frac{2E_b}{5}}$ $\quad S_4 = \langle S_{4l}(t), \Phi_l(t) \rangle = 3\sqrt{\frac{2E_b}{5}}$

Vector form: $S_1 = (-3\sqrt{\frac{2E_b}{5}}, 0)$ $S_2 = (-\sqrt{\frac{2E_b}{5}}, 0)$

$S_3 = (\sqrt{\frac{2E_b}{5}}, 0) \qquad S_4 = (-3\sqrt{\frac{2E_b}{5}}, 0)$

b. Derive the exact BER for 2-PAM and 4-PAM in terms of the signal-to-noise ratio (SNR) defined as $\boldsymbol{\gamma_b = \frac{E_b}{N_0}}$ when the Gray mapping is applied.

2PAM:

$$P_e = \int_{-\infty}^{0} P(s_2, r)dr + \int_{0}^{\infty} P(s_1, r)dr$$

$$= \int_{-\infty}^{0} P(r|s_2)P(s_2)dr + \int_{0}^{\infty} P(r|s_1)P(s_1)dr$$

$$= \frac{1}{2}\left(\int_{-\infty}^{0} \frac{1}{\sqrt{\pi N_0}} e^{-\frac{\left(r-\sqrt{\varepsilon_b}\right)^2}{N_0}} dr + \int_{0}^{\infty} \frac{1}{\sqrt{\pi N_0}} e^{-\frac{\left(r+\sqrt{\varepsilon_b}\right)^2}{N_0}} dr\right)$$

$$= Q\left(\sqrt{\frac{2\varepsilon_b}{N_0}}\right) = Q\left(\sqrt{\frac{2E_b}{N_0}}\right) = Q\left(\sqrt{2\boldsymbol{\gamma_b}}\right)$$

Since there are 2 signals, BER = SER=$Q\left(\sqrt{2\boldsymbol{\gamma_b}}\right)$.

4PAM:

$S_1 = (-3\sqrt{\frac{2E_b}{5}}, 0)$ $S_2 = (-\sqrt{\frac{2E_b}{5}}, 0)$ $\quad S_3 = (\sqrt{\frac{2E_b}{5}}, 0) \qquad S_4 = (3\sqrt{\frac{2E_b}{5}}, 0)$

Assume that $A = \sqrt{\frac{2E_b}{5}}$

Therefore, $S_1 = (-3A, 0)$ $S_2 = (-A,\ 0)$ $S_3 = (\mathrm{A},\ 0)$ $S_4 = (3\mathrm{A},\ 0)$

$$\mathrm{P}_{\mathrm{e}|\mathrm{S}_1} = \frac{1}{2}P_{S_2|S_1} + P_{S_3|S_1} + \frac{1}{2}P_{S_4|S_1}$$

$$P_{S_2|S_1} = \int_{-2A}^{0} P(s_1, r)dr = \int_{-2A}^{0} P(r|s_1)P(s_1)dr = \frac{1}{4}\int_{-2A}^{0} \frac{1}{\sqrt{\pi N_0}} e^{-\frac{(r+3A)^2}{N_0}} dr$$

$$= \frac{1}{4}\left[Q\left(A\sqrt{\frac{2}{N_0}}\right) - Q(3A\sqrt{\frac{2}{N_0}})\right]$$

$$P_{S_3|S_1} = \int_0^{2A} P(s_1, r)dr = \int_0^{2A} P(r|s_1)P(s_1)dr = \frac{1}{4}\int_0^{2A} \frac{1}{\sqrt{\pi N_0}} e^{-\frac{(r+3A)^2}{N_0}} dr$$

$$= \frac{1}{4}\left[Q\left(3A\sqrt{\frac{2}{N_0}}\right) - Q\left(5A\sqrt{\frac{2}{N_0}}\right)\right]$$

$$P_{S_4|S_1} = \int_{2A}^{\infty} P(s_1, r)dr = \int_{2A}^{\infty} P(r|s_1)P(s_1)dr = \frac{1}{4}\int_{2A}^{\infty} \frac{1}{\sqrt{\pi N_0}} e^{-\frac{(r+3A)^2}{N_0}} dr$$

$$= \frac{1}{4}\left[Q(5A\sqrt{\frac{2}{N_0}})\right]$$

$$\mathrm{P}_{\mathrm{e}|\mathrm{S}_1} = \frac{1}{2}P_{S_2|S_1} + P_{S_3|S_1} + \frac{1}{2}P_{S_4|S_1}$$

$$= \frac{1}{2}\frac{1}{4}\left[Q\left(A\sqrt{\frac{2}{N_0}}\right) - Q(3A\sqrt{\frac{2}{N_0}})\right] + \frac{1}{4}\left[Q\left(3A\sqrt{\frac{2}{N_0}}\right) - Q\left(5A\sqrt{\frac{2}{N_0}}\right)\right]$$

$$+ \frac{1}{2}\frac{1}{4}\left[Q(5A\sqrt{\frac{2}{N_0}})\right] = \frac{1}{8}\left[Q\left(A\sqrt{\frac{2}{N_0}}\right) + Q\left(3A\sqrt{\frac{2}{N_0}}\right) - Q(5A\sqrt{\frac{2}{N_0}})\right]$$

$$P_{e|S_2} = \frac{1}{2}P_{S_1|S_2} + \frac{1}{2}P_{S_3|S_2} + P_{S_4|S_2}$$

$$P_{S_1|S_2} = \int_{-\infty}^{-2A} P(s_2, r)dr = \int_{-\infty}^{-2A} P(r|s_2)P(s_2)dr = \frac{1}{4}\int_{-\infty}^{-2A} \frac{1}{\sqrt{\pi N_0}} e^{-\frac{(r+A)^2}{N_0}} dr$$

$$= \frac{1}{4}\left[Q\left(A\sqrt{\frac{2}{N_0}}\right)\right]$$

$$P_{S_3|S_2} = \int_0^{2A} P(s_2, r)dr = \int_0^{2A} P(r|s_2)P(s_2)dr = \frac{1}{4}\int_0^{2A} \frac{1}{\sqrt{\pi N_0}} e^{-\frac{(r+A)^2}{N_0}} dr$$

$$= \frac{1}{4}\left[Q\left(A\sqrt{\frac{2}{N_0}}\right) - Q\left(3A\sqrt{\frac{2}{N_0}}\right)\right]$$

$$P_{S_4|S_2} = \int_{2A}^{\infty} P(s_2, r)dr = \int_{2A}^{\infty} P(r|s_2)P(s_2)dr = \frac{1}{4}\int_{2A}^{\infty} \frac{1}{\sqrt{\pi N_0}} e^{-\frac{(r+A)^2}{N_0}} dr$$

$$= \frac{1}{4}\left[Q(3A\sqrt{\frac{2}{N_0}})\right]$$

$$P_{e|S_2} = \frac{1}{2}P_{S_1|S_2} + \frac{1}{2}P_{S_3|S_2} + P_{S_4|S_2}$$

$$= \frac{1}{2}\frac{1}{4}\left[Q\left(A\sqrt{\frac{2}{N_0}}\right)\right] + \frac{1}{2}\frac{1}{4}\left[Q\left(A\sqrt{\frac{2}{N_0}}\right) - Q\left(3A\sqrt{\frac{2}{N_0}}\right)\right]$$

$$+ \frac{1}{4}\left[Q(3A\sqrt{\frac{2}{N_0}})\right] = \frac{1}{8}\left[2Q\left(A\sqrt{\frac{2}{N_0}}\right) + Q\left(3A\sqrt{\frac{2}{N_0}}\right)\right]$$

S_1 and S_4 are symmetric, S_2 and S_3 are symmetric.

Therefore, $P_{e|S_1} = P_{e|S_4}, P_{e|S_2} = P_{e|S_3}$.

$$P_e = P_{e|S_1} + P_{e|S_2} + P_{e|S_3} + P_{e|S_4} = 2P_{e|S_1} + 2P_{e|S_2}$$

$$= \frac{1}{4}\left[3Q\left(A\sqrt{\frac{2}{N_0}}\right) + 2Q\left(3A\sqrt{\frac{2}{N_0}}\right) - Q\left(5A\sqrt{\frac{2}{N_0}}\right)\right]$$

As $A = \sqrt{\frac{2E_b}{5}}$ $\boldsymbol{\gamma_b} = \frac{\boldsymbol{E_b}}{\boldsymbol{N_0}}$ $P_e = \frac{1}{4}\left[3Q\left(2\sqrt{\frac{\gamma_b}{5}}\right) + 2Q\left(6\sqrt{\frac{\gamma_b}{5}}\right) - Q\left(10\sqrt{\frac{\gamma_b}{5}}\right)\right]$

c.Simulate 2-PAM and 4-PAM modulation and detection in baseband with AWGN to find the experimental BER using Jupyter notebook or Matlab. Please simulate until the BER reaches 10^{-6}. Plot the curve of BER vs. SNR. In this plot, the horizontal axis is SNR in dB-scale ($10\log_{10}$ SNR), and the vertical axis is BER in log-scale ($\log_{10}$ BER).

See picture in problem e

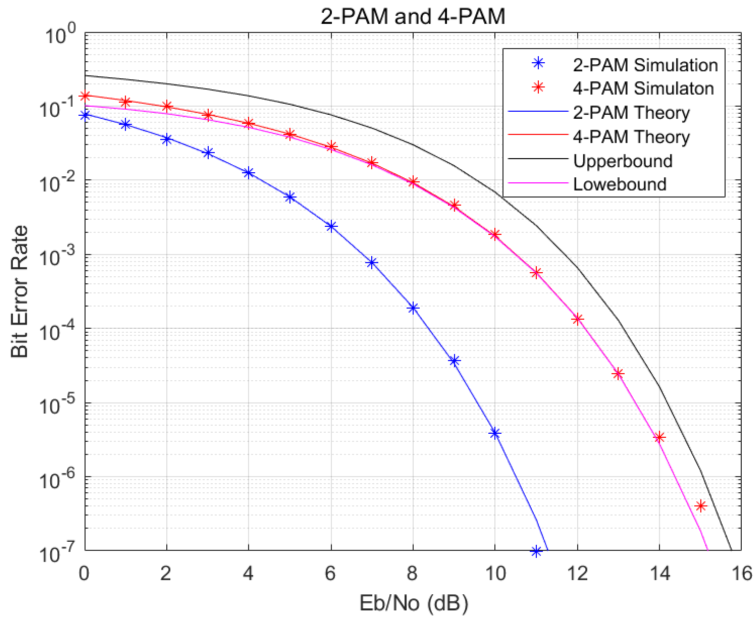
d.Derive an upper bound and a lower bound for the 4PAM BER (in terms of γ_b).

From the tutorial we have proven that $\frac{x}{\sqrt{2\pi}(1+x^2)}e^{-\frac{x^2}{2}} < Q(x) \leq \frac{1}{2}e^{-\frac{x^2}{2}}$

As $P_e = \frac{1}{4}\left[3Q\left(2\sqrt{\frac{\gamma_b}{5}}\right) + 2Q\left(6\sqrt{\frac{\gamma_b}{5}}\right) - Q\left(10\sqrt{\frac{\gamma_b}{5}}\right)\right]$

Replace Q(x) in inequality with $Q\left(2\sqrt{\frac{\gamma_b}{5}}\right) Q\left(6\sqrt{\frac{\gamma_b}{5}}\right) Q\left(10\sqrt{\frac{\gamma_b}{5}}\right)$ so we can get the upper bound and the lower bound

e. Plot the exact BER vs. SNR, the upper bound, and the lower bound in the same plot and comment on the results..



From the result, at the same signal-to-noise ratio, we can see that the BER of 2PAM is lower than that of 4PAM, indicating better modulation performance. And the bounds we derived meet the simulation requirements

Task 2

Assume all symbols are equiprobable. Denote the average bit energy by E_b. Derive and simulate the bit error rate (BER) for QPSK (4PSK) when the Gray mapping is applied.

a. Sketch the constellation of the QPSK.

b. Mark the symbol vector forms (in terms of E_b) and the corresponding Gray coding.

Assume $S_m(t) = Re\left[p(t)e^{j\left(\theta_0+\frac{2\pi(m-1)}{M}\right)}e^{j2\pi f_c t}\right] = p(t)\cos(2\pi f_c t + \theta_0 + \frac{2\pi(m-1)}{M})$

Denote $\varepsilon_p = \int_{-\infty}^{\infty} p(t)^2 dt \quad \rightarrow \quad \varepsilon_{avg} = \varepsilon_m = \frac{\varepsilon_p}{2} \qquad \varepsilon_{bavg} = \frac{\varepsilon_{avg}}{\log_2 M} = \frac{\varepsilon_p}{4} = E_b$

PAM has dimension two

$$\left\{\Phi_1(t) = \sqrt{\frac{2}{\varepsilon_p}}p(t)\cos(2\pi f_c t)\,, \Phi_2(t) = \sqrt{\frac{2}{\varepsilon_p}}p(t)\sin 2\pi f_c t\right\}$$

$$S_{11} = \langle S_1(t), \Phi_1(t)\rangle = \int_{-\infty}^{\infty} p(t)\cos(2\pi f_c t + \theta_0) * \sqrt{\frac{2}{\varepsilon_p}}p(t)\cos(2\pi f_c t)\,dt = \sqrt{\frac{\varepsilon_p}{2}}$$

$$= \sqrt{2E_b}$$

$$S_{12} = \langle S_1(t), \Phi_2(t)\rangle = 0$$

$$S_{21} = \langle S_2(t), \Phi_1(t)\rangle = 0 \qquad S_{22} = \langle S_2(t), \Phi_2(t)\rangle = \sqrt{2E_b}$$

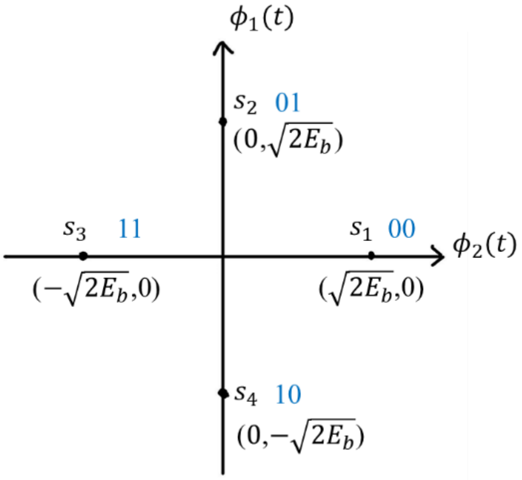
$$S_{31} = \langle S_3(t), \Phi_1(t)\rangle = -\sqrt{2E_b} \qquad S_{32} = \langle S_3(t), \Phi_2(t)\rangle = 0$$

$$S_{41} = \langle S_4(t), \Phi_1(t)\rangle = 0 \qquad S_{42} = \langle S_4(t), \Phi_2(t)\rangle = -\sqrt{2E_b}$$

Then the symbol vector forms of QPSK signals in terms of E_b and the corresponding Gray coding areϕ

signal	symbol vector form	Gray code
s1	$(\sqrt{2E_b},0)$	00
s2	$(0,\sqrt{2E_b})$	01
s3	$(-\sqrt{2E_b},0)$	11
s4	$(0,-\sqrt{2E_b})$	10

The constellation of the QPSK is shown as below.



c. **Derive the exact BER and express it in terms of the bit SNR $\boldsymbol{\gamma_b} = \frac{\boldsymbol{E_b}}{\boldsymbol{N_0}}$ when the Gray mapping is applied. Derive a upper bound and a lower bound for the QPSK BER in terms of SNR.**

For a 4-ary signaling in an AWGN channel, all symbols are equiprobable $\frac{1}{4}$ and bit SNR is $\boldsymbol{\gamma_b} = \frac{\boldsymbol{E_b}}{\boldsymbol{N_0}}$

The SER of the QPSK signals is

$$P_e = \frac{1}{M}\sum_{i=1}^{4} P(e|S_i) = P(e|S_i) = \sum_{\substack{1\le k\le M\\ k\neq 1}} \int_{D_k} P(\boldsymbol{r}|s_i)d\boldsymbol{r}$$

We can see s_1 is sent by detected as any S_k for any $k \neq m$

For pair-wise SER, $P_{m\to n} = Q(\sqrt{\frac{d_{mn}^2}{2N_0}})$, s_1 is affected by $s_{2,3,4}$, $d_{12} = d_{14} = 2\sqrt{E_b}, d_{13} = 2\sqrt{2E_b}$

$$P_{1\to 2} = P_{1\to 4} = Q\left(\sqrt{\frac{4E_b}{2N_0}}\right) = Q\left(\sqrt{\frac{2E_b}{N_0}}\right)$$

Then we can get BER of the QPSK signals

$$P_b = \frac{1}{M}\sum_{i=1}^{4} P(b|S_i) = P(b|S_i) = P(b|S_1)$$

$$= \frac{1}{2}\int_{D_2} P(\boldsymbol{r}|s_1)d\boldsymbol{r} + \int_{D_3} P(\boldsymbol{r}|s_1)d\boldsymbol{r} + \frac{1}{2}\int_{D_4} P(\boldsymbol{r}|s_1)d\boldsymbol{r}$$

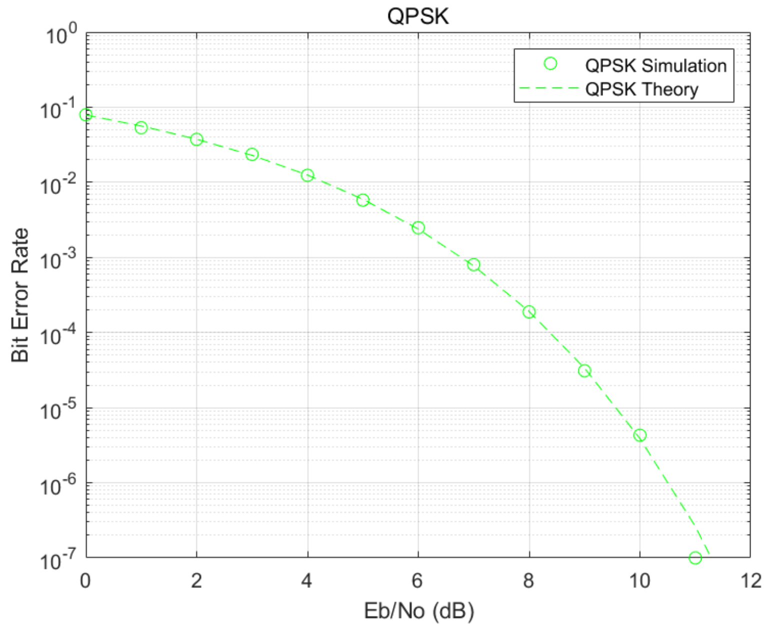
$$= \int_{D_2} P(\boldsymbol{r}|s_1)d\boldsymbol{r} + \int_{D_3} P(\boldsymbol{r}|s_1)d\boldsymbol{r} = P_{1\to 2} = Q\left(\sqrt{\frac{2E_b}{N_0}}\right)$$

Because $\boldsymbol{\gamma_b} = \frac{\boldsymbol{E_b}}{\boldsymbol{N_0}}$ $\quad P_b = Q(\sqrt{2\boldsymbol{\gamma_b}})$

d. **Simulate QPSK modulation and detection in baseband with AWGN to obtain experimental BER using Jupyter notebook/Python or Matlab. Please simulate until the BER reaches 10^{-6}. Plot the curve of BER vs. SNR. In this plot, the**

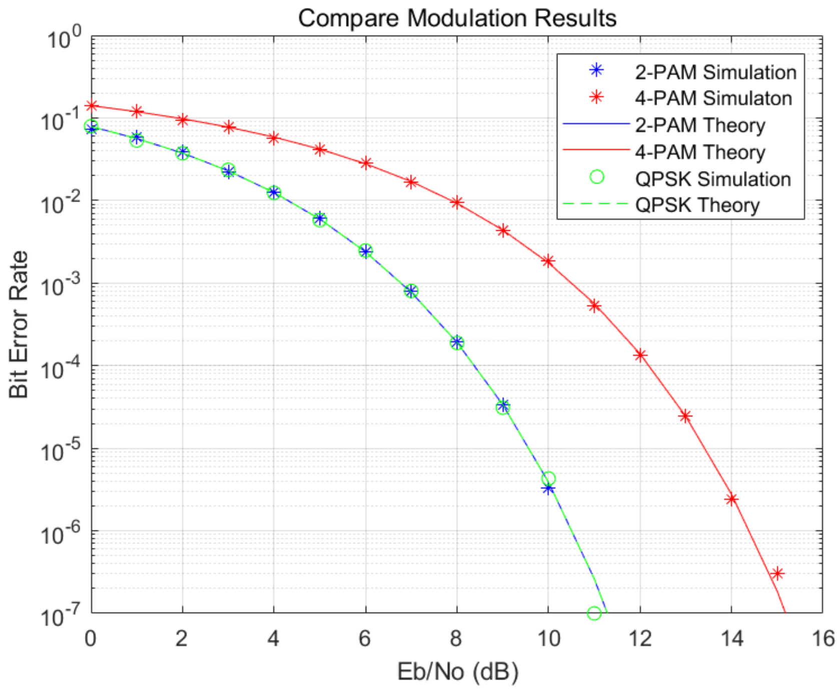
horizontal axis is SNR in dB-scale, and the vertical axis is BER in log-scale.

e. Plot the exact BER vs. SNR and comment on the result.



From the result, we can see BER decreases as the SNR gradually increases. Theoretical bit error rate is the same as actual bit error rate.

f. Compare the simulation results in Task 1 and Task 2.



From the result, the relationship of BER among the three modulation modes can be obtained at the same SNR:

BER: $4PAM > QPSK = 2PAM$

In other words, at the same BER, 4PAM requires a higher SNR, while 2PAM and QPSK have higher efficiency.